

Week2 Homework

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1. Graph Basics

a. Connected components: (a, d, f), (b, e), (c, g).

Cycle: a-d-f-a.

The minimum number of new edges required to make the graph with k connected components connected: $k - 1$. $k - 1$ is valid: You can link component 1 to component $[2, k]$. $< k - 1$ is impossible: If we can connect with $k' < k - 1$ edges, then we delete a vertex with $\deg = 1$ (if this kind of vertex is not exist, then there is $2k/2 = k$ edges), then we can connect $k - 1$ component with $k' - 1$ edges. Keep doing this, we can connect $k - k' > 1$ components with 0 edge.

One possible set for G is: $(a, b), (a, c)$.

b. T is a spanning tree for H . c to f : $c - b - d - e - f$. Cycle: $c - b - d - e - f - c$. Removable edge: $(c, b), (b, d), (d, e), (e, f)$.

2. Algorithm and Time complexity

a.1. $[1, 3, 4, 6, 8, 9, 10]$, 6.

a.2. $T_{median}(2^{30}) = 20 \cdot 2^{30}$, $T_{sort}(2^{30}) = 60 \cdot 2^{30}$, so median is fewer by factor $1/3$. We have $T_{median}(2^{10}) = T_{sort}(2^{10})$.

b. Binary Heap: $O((n + m) \log n)$. Fibonacci Heap: $O(n \log n + m)$.

c. Prim: $O(n \log n + m)$. When $m = \Theta(n)$, Prim is $O(n \log n)$ and Yao is $O(n \log \log n)$, so Yao is faster. When $m = \Theta(n \log n)$, Prim is $O(n \log n)$ and Yao is $O(n \log n \log \log n)$, so Prim is faster.

3. Shortest Path

a. Finalize order: S, B, A, D, C, E, F, T. Shortest: S-A-C-E-T, 1300m.

b. Finalize order: S, A, C, E, T. Total: 5 vertices.

c. Let $h'(v) = 0.36h(v)$ representing the remaining time if use speed $20km/h$ and go straightly. Since you can never go faster than $20km/h$, and go curvly is worse than go straightly, the h' is admissible.

Finalize order: S, A, C, B, D, F, T. Fastest: S-B-D-F-T, 612s.

Declaration of AI Use

I used AI to translate part of words and search for Yao's algorithm.